Chronicle
Video-free memory (VFM) architecture

Technical note — 27 October 2026

Branch `experiment/video-free-memory` — main UI + VLM-first answers

One-sentence summary. Chronicle compiles a video episode *once* into a compact, timestamped memory — VLM captions, a salience-coalesced event ledger, a few WebP evidence frames and an LLM-ready pack — then answers questions from that memory in microseconds, with no video and no video model in the query path. The zine dashboard from `main` is the UI; the VLM memory is the answer engine.

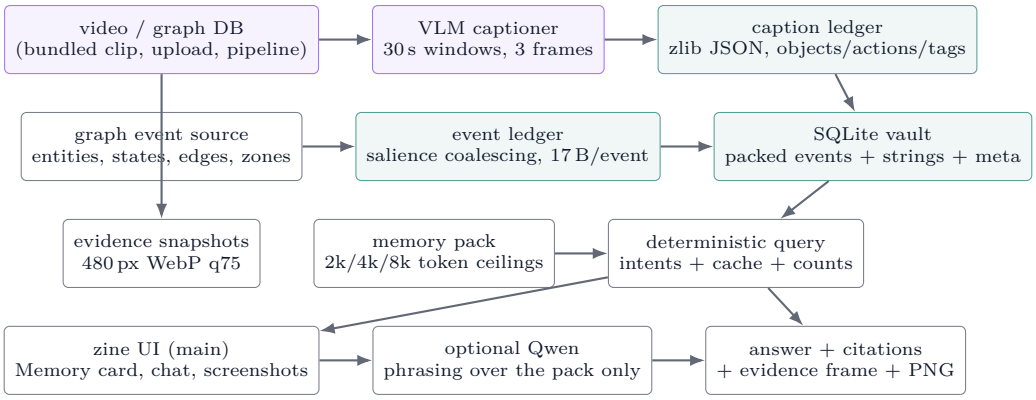
1 What is different from the graph-only dashboard

The `main` dashboard replays a live graph: entities, state transitions, spatial edges and zones. That is excellent for visual inspection, but it has two limitations for question answering:

- It only knows what the graph recorded. Replay clips often have **state/edge events but no VLM interaction table**, so "what happened around 10 s?" had no semantic answer.
- Labels can arrive late. If an entity row is emitted at `last_seen`, the first half of the episode shows anonymous `Track` rows.

VFM adds the layer the graph does not keep: **dense natural-language captions** at 30 s resolution, a compact event ledger derived from the graph, evidence snapshots, and a retrieval engine whose answers are deterministic. The graph remains the index; the captions and events are the recallable memory.

2 Architecture at a glance



The two inputs are compiled in parallel and joined in the vault. Query time reads only the vault and the pack. The source video is retained for playback and screenshot capture, but is never required to answer a question.

3 What is saved now (more than the graph)

The graph is still saved. VFM adds four new artefacts. All of them are derived once, compressed, and stored next to the source run; the video itself is optional at query time.

Artefact	What it contains	Demo	Walking
Graph DB	entities, states, edges, zones, observations (source of truth)	168 KB	254 KB
Packed vault	struct-of-arrays events, string dictionary, metadata, answer cache	28.7 KB	31.7 KB
Caption ledger	30 s VLM windows: caption, objects, actions, zone, tags, 3 frame captions	1 window	2 windows
Event ledger	salience-coalesced state/edge/interaction events	8 events	11 events
Evidence	480 px WebP stills, ranked by interaction / crop / zone keyframe	8 frames	8 frames
Memory pack	verified answer + ranked captions/events + evidence refs + rules	2.4 KB	3.0 KB
Report	single-file offline HTML with timeline, captions, query box	366 KB	410 KB
Whole job dir	vault + ledger + evidence + report + pack + zip	**988 KB**	**1.3 MB**

The report and zip duplicate bytes for convenience; the *at-rest memory* is the vault + ledger + evidence. On a 60-minute synthetic episode (2,474 events, 184 caption windows, 40 snapshots) the vault is 142 KB and the whole memory stays far below the source video.

3.1 Why this is more information

- **Semantics.** The graph knows `person VISIBLE_MOVING`; the caption knows "a person walks on a sidewalk holding a cup of coffee". The second is what a user asks for.
- **Coverage.** Captions cover the whole episode, including segments where the tracker emitted no interaction row.
- **Recall by time.** Caption windows are timestamped, so "around 10 s" is a range query, not a similarity search over one frame.
- **Evidence.** Answers point to stored stills, so the UI can show the frame and seek the video without re-ingesting.
- **Counts.** Compile-time observations also yield track counts, peak simultaneous visibility, and cross-label duplicate merges (Section 6).

3.2 Design lineage

The design deliberately borrows one idea from each line of work: caption-first episodic QA [4], time as a first-class key [5], query-agnostic high-recall memory [6], egocentric entity/event schemas [3, 2, 1], graph retrieval [11], tiered sensory/short/long-term memory [10], event segmentation [9], bounded retention [8], text-first / frames-on-demand [7, 16], semantic answer caching [13], and personal-memory retrieval [12]. The model-native memory work (Titans, MemoryLLM) [14, 15] is a north star, not a dependency.

4 Compile pipeline

1. **Snapshot the source DB.** SQLite's backup API makes a consistent copy while the ingest worker may still be writing.
2. **Caption once.** Split the video into 30 s windows; sample three frames per window; caption each frame with a VLM (`qwen2.5vl:3b` by default) and synthesise a window caption. Store zlib-compressed JSON with objects, actions, zone and tags.
3. **Build the event ledger.** Read entities, state transitions, anchors and edges; coalesce visibility noise and repeated touches into salient events. Pack them struct-of-arrays at roughly 17 bytes/event.
4. **Build the vault.** Write 1 KB SQLite pages, the string dictionary, packed events, captions and metadata (including per-label count summaries).
5. **Extract evidence.** Save a bounded set of 480 px WebP q75 stills (interaction strips, object crops, zone keyframes). No full-resolution frames are retained.
6. **Render the report.** Write a single-file HTML with the timeline, captions and a client-side query box; embed evidence as data URLs.
7. **Write the pack.** Produce an LLM-ready Markdown/JSON pack. Budgets of 2k/4k/8k tokens keep the verified answer and rules, then add windows/events until the ceiling.
8. **Zip and manifest.** Expose `vault.db`, `memory.html`, `ledger.json`, `memory_pack.md/json`, CSVs and `evidence/`.

The compile is the only expensive step (seconds to tens of seconds, dominated by VLM calls). In the UI it runs *once per clip, in the background*, on first view. After that, opening the clip or asking a question does not touch the video.

5 Query path and latency

Deterministic vault. The query engine resolves an intent (where/when/around/before/after/ count/inventory/yes-no/keyword), looks up entities and events by time, and returns an answer with citations and an evidence reference. A canonical answer cache makes paraphrases hit the same entry. No LLM, no video, no frame extraction in this path.

Qwen (optional). The pack is passed to a local text model for phrasing (`qwen2.5:7b-instruct`). The model never sees frames; it sees the verified structured answer, timestamped captions, events and evidence references. It is an optional polish layer, not a dependency.

Path	Accuracy (30 questions)	p50	Video at answer time
Deterministic VFM vault	**30/30**	**0.05–0.06 ms**	no
Qwen 7B from memory pack	**30/30**	1.07 s	no
Original graph + Qwen 7B	14/30	0.85 s	yes
9-frame contact sheet + Qwen2.5-VL	7/30	4.46 s	yes (19.24 MB)

The 30-question set is an internal regression suite, not a public benchmark. It is useful because it covers location, temporal, count, negation, before/after and paraphrase questions on the same episodes.

6 Counts: tracks, visibility and duplicates

Count questions need more than one number. At compile time we compute, per label:

Field	Meaning
`tracks`	raw number of tracker ids with that label (upper bound)
`peak`	max same-label tracks present in one observation frame (lower bound)
`physical`	tracks after removing cross-label duplicate detections and one-frame transients
`duplicates`	tracks that were near-identical to a track under another label
`merged_into`	which label absorbed those duplicates

A duplicate is accepted only when two different-label boxes share the same frame with IoU ≥ 0.90 and centre distance ≤ 5 px, and one label has more total observations. This is deliberately conservative: same-label co-visible boxes on cook have max IoU ≈ 0.01, so they are *not* merged.

On the cook clip this turns four "bowl" tracks into **zero physical bowls**: two of the plates were emitted under both `bowl` and `plate` (IoU 0.91–0.99), and a fourth track was a one-frame detection. The plates resolve to two physical plates. On walking, two car tracks are one object at a time, so the answer is a range (1–2), not a false single count.

7 Dashboard integration

The zine mission control from `main` is the default UI (`./chronicle.sh start`, port 8080 in main, 8082 in this worktree). VFM is added behind it, not as a second uploader:

- **Auto-compile.** On first view of a clip with a graph DB, the vault builds in the background. The Memory card shows progress, then report and downloads.
- **VLM-first chat.** Once ready, the existing chat routes to the vault (`provider: vfm-memory`) and answers in microseconds. If not ready, a state/edge fallback still answers "around t" instead of claiming no interactions exist.
- **Screenshot at answer time.** Every grounded answer gets a frame chip that seeks to the citation, captures the exact frame, stamps clip + timestamp, and opens a lightbox.
- **Fullscreen.** Stage toolbar `S` (screenshot) and `F` (fullscreen); the screenshot lightbox has its own fullscreen button.
- **Downloads.** `vault.db`, `memory.html`, `memory_pack.md`, `memory_pack.json`, `memory.zip`.

8 Honest limits

- `physical` is a conservative estimate, not hand-labelled ground truth for every episode. Cook plates/bowls were checked visually; walking cars were not.
- Counts are label-dependent. An armchair labelled `sofa`, or a bowl never co-detected with a plate, still contributes to the wrong label.
- Caption-first uploads (no graph DB) have no observations, so they keep caption-coverage wording and must not claim physical counts.
- The 30-question suite is a regression test, not a public benchmark. CapMem/EgoSchema/Ego4D would test questions we did not design.
- The VLM can hallucinate detail; captions need spot checks. Evidence stills and timestamped captions make the claim auditable.

9 References and papers used

The following works shaped the design. Citation numbers appear in the text where the idea is used.

References

[1] R4DSG. *Relative 4D scene-graph memory for egocentric video.* ACM MM 2026. arXiv:2608.11017. Anchor-relative, video-free scene memory; validates compact queryable entries.

[2] VL-KnG. *Persistent spatio-temporal knowledge graph from monocular egocentric video.* 2025/2026. arXiv:2510.01483. Closest architectural cousin: entities + events + time, query latency decoupled from video length.

[3] EgoGraph. *Training-free temporal knowledge graph for egocentric memory.* 2026. arXiv:2602.23709. Egocentric schema of people, objects, locations and events; temporal modelling across entities.

[4] CapMem. *Episodic memory video caption QA.* EMNLP 2026. arXiv:2609.17688. Captions-only QA beats direct VideoQA on long videos for most models; motivates the 30 s caption ledger and the caption-window study.

[5] EgoCITE. *Dense visual captions + speech transcripts as memory entries.* 2026. arXiv:2608.12627v2. Time is a first-class retrieval key; motivates timestamped windows and temporal filters.

[6] MERIT. *Query-agnostic memory build; episodic multi-key representation.* ECCV 2026. arXiv:2608.07663. Build high-recall memory before knowing the question; compose/expand only at query time.

[7] FocusGraph. *Scene-graph LLM selector + sparse keyframes.* 2026. arXiv:2603.04349. Text-first, pixels-second routing; supports the evidence-on-demand ladder.

[8] VideoMem. *Adaptive global memory buffer for ultra-long video.* 2025/2026. arXiv:2512.04540. Compression and retention policy inspiration.

[9] EM-LLM. *Episodic events from Bayesian surprise + graph boundary refinement.* ICLR 2025. ICLR proceedings. Event segmentation and similarity + temporal-contiguity retrieval.

[10] LightMem. *Sensory → short-term → long-term memory; offline consolidation.* ICLR 2026. arXiv:2510.18866. Tiering and the offline/online split; ingest is our offline phase.

[11] HippoRAG. *Knowledge graph + Personalized PageRank for long-term memory.* NeurIPS 2024. arXiv:2405.14831. Graph retrieval as a cheap alternative to iterative VLM calls.

[12] EgoMemory / EgoRetriever. *Personal memory bank + reflective CoT; training-free retrieval over long egocentric video.* ACL Findings 2026. ACL anthology. Personalisation and intent inference over a memory bank.

[13] Continuous Semantic Caching. *Principled semantic cache reuse under uncertainty.* 2026. CMU LIONS. The answer cache is a memory tier; paraphrases can reuse grounded answers.

[14] Titans. *Neural long-term memory that learns at test time.* 2024/2025. arXiv:2501.00663. North star: memory in weights, not only a database.

[15] MemoryLLM. *Fixed-size latent memory pool inside a transformer.* ICML 2024. arXiv:2402.04624. Alternative route to model-native episode memory; not on the demo critical path.

[16] Caption-once, Frames-on-Demand. *Cache captions once; fetch frames only when confidence is low.* EMNLP 2026. ACL anthology. Exactly the query-time cost model: text first, bounded pixels on demand.

Appendix: commands

```
# Start the merged zine UI (main UI + VFM memory)
cd /home/g1/projects/chronicle-vfm
./chronicle.sh start            # this worktree defaults to :8082
./chronicle.sh status
./chronicle.sh stop

# Full verification
./chronicle.sh test

# Compile a memory directly (no UI)
python prototype/compile.py --source-db data/real/cook/chronicle.db --out /tmp/cook.db
python prototype/attach_ledger.py --vault /tmp/cook.db --ledger /tmp/cook_ledger_30.json
python prototype/query.py --db /tmp/cook.db "how many bowls?"
python prototype/report.py --db /tmp/cook.db --out /tmp/cook.html --data-root /tmp

# 2k / 4k / 8k pack ceilings
python prototype/memory_pack.py --db /tmp/cook.db --list-budgets
```